



World's first 'roll-to-roll' manufactured OLED from GE

supplies 1500 lumens of light output," notes Thienken.

He adds, "High-power LEDs will continue to offer increased levels of performance. We believe other developments will enable lighting customers to simplify their designs and manufacturing and also reduce the cost of LED lighting products."

Organic might be the answer

It is evident that almost every company is focusing on two things as far as research goes: increasing the performance and decreasing the cost.

If manufacturers are able to figure out the right technique, organic LEDs made using inexpensive polymers might be the answer. However, at present, OLEDs are known to have a lower efficiency and lifespan than inorganic crystalline LEDs. Despite that, GE seems to have found out how to produce nice, bright OLEDs.

Way back in 2004, the company demonstrated an OLED device that was fully functional as a 61x61cm² panel, and produced 1200 lumens of light with efficiency on par with today's incandescent bulb technology. Since then, GE has more than doubled the level of OLED efficiency using device architectures that are

scalable to a large area and can be produced cost-effectively. The efforts to increase the efficiency and performance of OLED lighting have coincided with the development of a low-cost, roll-to-roll process for manufacturing these devices.

In 2008, they demonstrated the first roll-to-roll manufactured OLED lighting devices. This can be considered a key step toward making OLEDs and other high-performance organic electronics products at a very low cost. What is more, this effort was led by an Indian, Anil Duggal. By next year, we might see OLED lighting products from GE.

Quantum leaps ahead

Michael Bowers of Vanderbilt University, Nashville, USA, has developed an experimental technique that involves coating a blue LED with quantum dots that glow white in response to the blue light from the LED. The mellow yellow

light produced by this method is quite similar to that of our incandescent bulb.

According to the Wikipedia, quantum dots are semiconductor nanocrystals that possess unique optical properties. Their emission colour can be tuned from the visible throughout the infrared spectrum. This allows quantum-dot LEDs to create almost any colour on the CIE diagram, providing more colour options and better-colour-rendering white LEDs. Quantum-dot LEDs are available in the same package types as traditional phosphor-based LEDs. Apparently, the Nanoco Group and a Japanese company have also tied up to design, develop and produce these quantum dots.

In all, it is quite a promising scene. There is a demand and a supply for LEDs, for niche as well as mass purposes. Manufacturers are hooked onto so many R&D programmes, to reduce cost, increase performance, improve the colour rendering and uniformity of light, achieve constant brightness throughout the life of the product, enhance the packaging, customise the LED components to suit specific applications, etc in order to cater to niche industries as well as mass adoption.

"At this point of time, LEDs are thriving to lower energy demand but in the future LED luminaires with superior aesthetics, automatic controls, thermal management and lower costs will lead the way. These will further promote wider applicability, mass availability and greater affordability," says Cherred. "We believe that the magic differentiator will be a consortium approach of system integrators, LED manufacturers and electronics manufacturers, which will then lead to market consolidation and leadership. This will create better reference designs, superior products and lower total costs of ownership," he adds. ●

Today's white LED fixtures average around 46 lm/W, which is much higher than what incandescent bulbs and fluorescent bulbs have been able to manage till date

The author is a freelance writer based in Bengaluru. She writes on a variety of topics, her favourites being technology, cuisine, and life